Project: Worked examples
Title: Member checks (EN 1993-1-1)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Resistance checks of a steel member of class 1, 2 or 3 cross-section: tension (6.2.3), compression (6.2.4) and flexural buckling about y and z (6.3.1), bending about y and z (6.2.5), shear with the high-shear reduction (6.2.6, 6.2.8), the linear cross-section interaction (6.2.1(7)) and a simplified member interaction (6.3.3 with user-entered k factors). Lateral-torsional buckling and class 4 sections are outside this sheet.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Axial plastic resistance | $N_{pl,Rd} = A\, f_y/\gamma_{M0}$ |
| Bending resistance, plastic modulus for class 1–2 and elastic for class 3 | $M_{c,Rd} = W\, f_y/\gamma_{M0}$ |
| Shear resistance | $V_{pl,Rd} = A_v\, f_y/(\sqrt{3}\,\gamma_{M0})$ |
| Bending resistance reduced by high shear, when $V_{Ed} > 0.5\, V_{pl,Rd}$ | $\rho = (2V_{Ed}/V_{pl,Rd} - 1)^2$. Class 1–2 I-section with h, $t_w$, $t_f$ entered: $M_{y,V,Rd} = (W_{pl,y} - \rho\, A_w^2/(4\, t_w))\, f_y/\gamma_{M0}$, $A_w = (h - 2t_f)\, t_w$, and $M_{z,c,Rd}$ is not reduced; otherwise the whole resistance × $(1 - \rho)$ |
| Elastic critical load and non-dimensional slenderness | $N_{cr} = \pi^2 EI/L_{cr}^2$; $\bar{\lambda} = \sqrt{(A\, f_y/N_{cr})}$ |
| Buckling reduction factor and buckling resistance | $\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$; $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$; $N_{b,Rd} = \chi\, A\, f_y/\gamma_{M1}$ |
| Cross-section check, linear interaction | $N_{Ed}/N_{Rd} + M_{y,Ed}/M_{y,Rd} + M_{z,Ed}/M_{z,Rd} \le 1$ |
| Member check, axial force and bending with buckling (6.61, 6.62) | $N_{Ed}/N_{b,Rd,min} + k_{yy}\, M_{y,Ed}/(W_y\, f_y/\gamma_{M1}) + k_{zz}$ $M_{z,Ed}/(W_z\, f_y/\gamma_{M1}) \le 1$, the characteristic bending resistance $M_{Rk}$ over $\gamma_{M1}$ without the shear reduction |

Basis: Tension[1](6.2.3); Compression[1](6.2.4); Bending moment[1](6.2.5); Shear[1](6.2.6); Bending and shear[1](6.2.8); Linear summation of utilisations[1](6.2.1(7)); Uniform members in compression: buckling resistance, buckling curves[1](6.3.1.1, 6.3.1.2); Uniform members in bending and axial compression[1](6.3.3); Imperfection factors for the buckling curves[2](§6.3.1).

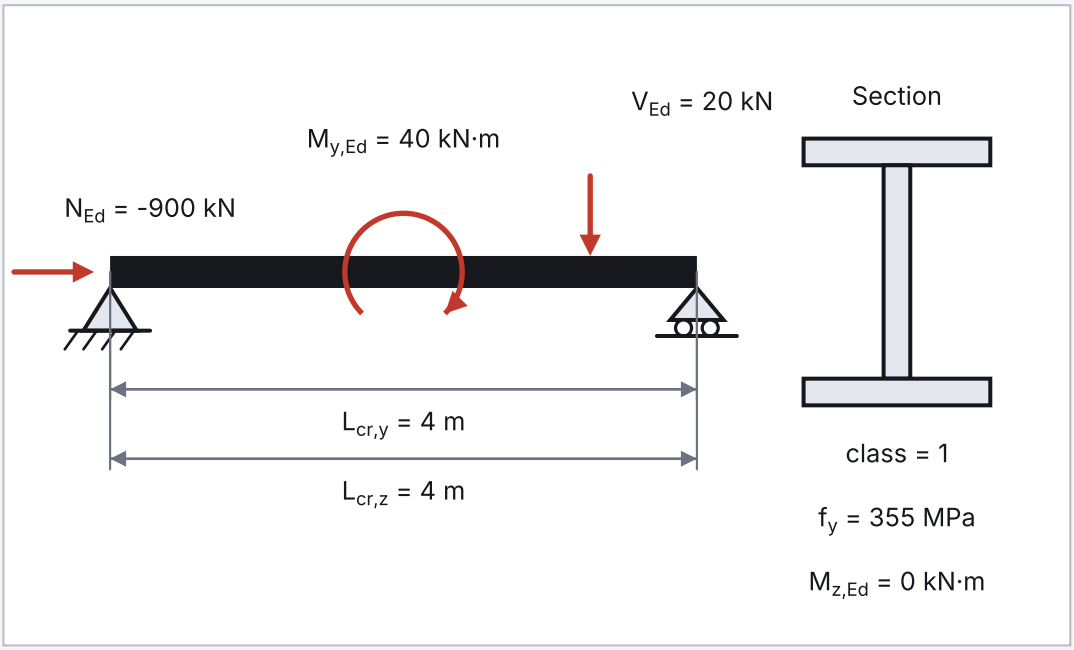



Figure 1 — A steel member under axial force, bending and shear, its buckling lengths about both axes, and its section. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $N_{Ed}$ | Design axial force, TENSION positive and compression negative (the opposite sign to S6's $N_{Ed}$, which is compression positive; S2's $N_{e1}$ is tension positive) | -900 | kN | reference: S2 frame sheet, column 1, ULS | -100000 … 100000 kN |
| $M_{y,Ed}$ | Design moment about y-y (major) | 40 | kN·m | reference: S2 frame sheet, column 1, ULS | 0 … 100000 kN·m |
| $M_{z,Ed}$ | Design moment about z-z (minor) | 0 | kN·m | reference: S2 frame sheet | 0 … 100000 kN·m |
| $V_{Ed}$ | Design shear force | 20 | kN | reference: S2 frame sheet | 0 … 100000 kN |
| $f_y$ | Yield strength (for the thickness) | 355 | MPa | spec: BS EN 10025-2 S355, t ≤ 16 mm | 200 … 700 MPa |
| E | Modulus of elasticity | 210 | GPa | spec: BS EN 1993-1-1 3.2.6 | 190 … 220 GPa |
| $\gamma_{M0}$ | Partial factor $\gamma_{M0}$ (cross-section resistance) | 1 | | spec: UK NA to BS EN 1993-1-1 NA.2.15 | 1 … 1.25 |
| $\gamma_{M1}$ | Partial factor $\gamma_{M1}$ (member buckling) | 1 | | spec: UK NA to BS EN 1993-1-1 NA.2.15 | 1 … 1.25 |
| $section_{class}$ | Cross-section class | 1 | | assumption: UC 203×203×46 S355 is class 1 in compression (Blue Book). | any |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| A | Area A | 58.7 | $cm^2$ | reference: S1 section sheet / Blue Book UC 203×203×46 | 1 … 100000 $cm^2$ |
| $W_{pl,y}$ | $W_{pl,y}$ | 497 | $cm^3$ | reference: S1 section sheet / Blue Book | 0.1 … 10000000 $cm^3$ |
| $W_{el,y}$ | $W_{el,y}$ | 450 | $cm^3$ | reference: S1 section sheet / Blue Book | 0.1 … 10000000 $cm^3$ |
| $W_{pl,z}$ | $W_{pl,z}$ | 231 | $cm^3$ | reference: S1 section sheet / Blue Book | 0.1 … 10000000 $cm^3$ |
| $W_{el,z}$ | $W_{el,z}$ | 152 | $cm^3$ | reference: S1 section sheet / Blue Book | 0.1 … 10000000 $cm^3$ |
| $I_y$ | $I_y$ | 4568 | $cm^4$ | reference: S1 section sheet / Blue Book | 0.1 … 1000000000 $cm^4$ |
| $I_z$ | $I_z$ | 1548 | $cm^4$ | reference: S1 section sheet / Blue Book | 0.1 … 1000000000 $cm^4$ |
| h | Depth h (shear-area helper; leave blank when $A_v$ is given) | 203.2 | mm | spec: BS EN 10365 UC 203×203×46 | 1 … 5000 mm |
| b | Width b (helper) | 203.6 | mm | spec: BS EN 10365 UC 203×203×46 | 1 … 5000 mm |
| $t_w$ | Web thickness (helper) | 7.2 | mm | spec: BS EN 10365 UC 203×203×46 | 0.1 … 500 mm |
| $t_f$ | Flange thickness (helper) | 11 | mm | spec: BS EN 10365 UC 203×203×46 | 0.1 … 500 mm |
| r | Root radius (helper) | 10.2 | mm | spec: BS EN 10365 UC 203×203×46 | 0 … 500 mm |
| η | Shear-area factor η | 1 | | spec: UK NA to BS EN 1993-1-5 NA.2.4 | 1 … 1.2 |
| $L_{cr,y}$ | Buckling length about y-y | 4 | m | assumption: Pinned both ends: L_cr = L = 4 m. | 0.1 … 100 m |
| $L_{cr,z}$ | Buckling length about z-z | 4 | m | assumption: Pinned both ends: L_cr = L = 4 m. | 0.1 … 100 m |
| $curve_y$ | Buckling curve about y-y | b | | spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, y-y | any |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $curve_z$ | Buckling curve about z-z | c | | spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, z-z | any |
| $k_{yy}$ | Interaction factor $k_{yy}$ (member check) | 1 | | assumption: Simplified interaction: k_yy = 1.0. | 0.5 … 2 |
| $k_{zz}$ | Interaction factor $k_{zz}$ (member check) | 1 | | assumption: Simplified interaction: k_zz = 1.0. | 0.5 … 2 |

## 3. Assumptions

A1. The cross-section class is as stated by the user (1, 2 or 3); class 4 effective properties are not derived here. (6.2.2.5 effective widths are outside this sheet.)
A2. Lateral-torsional buckling is not checked: the compression flange is assumed restrained, or LTB is checked elsewhere. (6.3.2 not implemented.)
A3. The member interaction uses the user's $k_{yy}$ and $k_{zz}$ (default 1.0) with the smaller flexural-buckling reduction on the axial term; the Annex A/B interaction factors are not derived. (6.3.3 simplified.)
A4. Imperfection factors per curve are the textbook values and may be overridden; partial factors default to 1.0 (UK National Annex) and may be overridden. (User-editable with provenance.)
A5. $section_{class}$ (Cross-section class) is taken as 1. (UC 203×203×46 S355 is class 1 in compression (Blue Book).)
A6. $L_{cr,y}$ (Buckling length about y-y) is taken as 4 m. (Pinned both ends: $L_{cr}$ = L = 4 m.)
A7. $L_{cr,z}$ (Buckling length about z-z) is taken as 4 m. (Pinned both ends: $L_{cr}$ = L = 4 m.)
A8. $k_{yy}$ (Interaction factor $k_{yy}$ (member check)) is taken as 1. (Simplified interaction: $k_{yy}$ = 1.0.)
A9. $k_{zz}$ (Interaction factor $k_{zz}$ (member check)) is taken as 1. (Simplified interaction: $k_{zz}$ = 1.0.)

## 4. Working

[27] $r_h = r$
= 10.2 mm
= 10.2 mm
Root radius used by the helper

[28] $A_{v,rolled} = A - 2 * b * t_f + (t_w + 2 * r_h) * t_f$ [1](6.2.6(3))
= 58.7 cm² − 2 × 203.6 mm × 11 mm + (7.2 mm + 2 × 10.2 mm) × 11 mm
= 16.944 cm²
Rolled I shear area, load parallel to the web

[29] $A_{v,min} = \eta * (h - 2 * t_f) * t_w$ [1](6.2.6(3))
= 1 × (203.2 mm − 2 × 11 mm) × 7.2 mm
= 13.046 cm²
Lower bound $\eta\ h_w\ t_w$

[30] $A_v = \max(A_{v,rolled}, A_{v,min})$
= max(16.944 cm², 13.046 cm²)
= 16.944 cm²
Shear area

[31] $N_{pl,Rd} = A * f_y / \gamma_{M0}$ [1](6.2.3(2), 6.2.4(2))
= 58.7 cm² × 355 MPa / 1
= 2083.8 kN
Plastic axial resistance

[32] $W_y = W_{pl,y}$ [1](6.2.5(2))
= 497 cm³
= 497 cm³

Section modulus about y (plastic, class 1–2)

[33] $W_z = W_{pl,z}$ [1](6.2.5(2))

$= 231\ cm^3$

$= 231\ cm^3$

Section modulus about z (plastic)

[34] $V_{pl,Rd} = A_v * f_y / (\sqrt{3} * \gamma_{M0})$ [1](6.2.6(2))

$= 16.944\ cm^2 \times 355\ MPa / (\sqrt{3} \times 1)$

$= 347.28\ kN$

Plastic shear resistance

[35] $u_V = V_{Ed} / V_{pl,Rd}$

$= 20\ kN / 347.28\ kN$

$= 0.05759$

Shear utilisation

[36] $\rho = 0$ [1](6.2.8(2))

$= 0$

Shear reduction factor ($V_{Ed} \le 0.5\ V_{pl,Rd}$)

[37] $M_{y,c,Rd} = (1 - \rho) * W_y * f_y / \gamma_{M0}$ [1](6.2.5(2), 6.2.8)

$= (1 - 0) \times 497\ cm^3 \times 355\ MPa / 1$

$= 176.44\ kN \cdot m$

Bending resistance about y

[38] $M_{z,c,Rd} = (1 - \rho) * W_z * f_y / \gamma_{M0}$ [1](6.2.5(2))

$= (1 - 0) \times 231\ cm^3 \times 355\ MPa / 1$

$= 82.005\ kN \cdot m$

Bending resistance about z

[39] $u_{M,y} = M_{y,Ed} / M_{y,c,Rd}$

$= 40\ kN \cdot m / 176.44\ kN \cdot m$

$= 0.22671$

Bending utilisation, y

[40] $u_{M,z} = M_{z,Ed} / M_{z,c,Rd}$

$= 0\ kN \cdot m / 82.005\ kN \cdot m$

$= 0$

Bending utilisation, z

[41] $u_N = \mathrm{abs}(N_{Ed}) / N_{pl,Rd}$

$= \mathrm{abs}(-900\ kN) / 2083.8\ kN$

$= 0.43189$

Compression (cross-section) utilisation

[42] $u_{section} = u_N + u_{M,y} + u_{M,z}$ [1](6.2.1(7))

$= 0.43189 + 0.22671 + 0$

$= 0.65861$

Linear cross-section interaction

[45] $\alpha_y$ = imperfection factor, curve b [1](6.3.1.2)

= Designers' Guide §6.3.1

$= 0.34$

Imperfection factor about y

[46] $\alpha_z$ = imperfection factor, curve c [1](6.3.1.2)

= Designers' Guide §6.3.1

$= 0.49$

Imperfection factor about z

[47] $N_{cr,y} = \pi^2 * E * I_y / L_{cr,y}^2$ [1](6.3.1.2(1))

$= \pi^2 \times 210\ GPa \times 4568\ cm^4 / (4\ m)^2$

$= 5917.3\ kN$

Elastic critical force about y

[48] $\lambda_y = \sqrt{A * f_y / N_{cr,y}}$ [1](6.3.1.2(1))

$= \sqrt{58.7\ cm^2 \times 355\ MPa / 5917.3\ kN}$

$= 0.59343$

Non-dimensional slenderness about y

[49] $\Phi_y = 0.5 * (1 + \alpha_y * (\lambda_y - 0.2) + \lambda_y^2)$ [1](6.3.1.2(1))

$= 0.5 \times (1 + 0.34 \times (0.59343 - 0.2) + 0.59343^2)$

$= 0.74296$

Φ about y

[50] $\chi_y = \min(1, 1 / (\Phi_y + \sqrt{\Phi_y^2 - \lambda_y^2}))$ [1](6.3.1.2(1))

$= \min(1, 1 / (0.74296 + \sqrt{0.74296^2 - 0.59343^2}))$

$= 0.84034$

Reduction factor χ about y

[51] $N_{b,y,Rd} = \chi_y * A * f_y / \gamma_{M1}$ [1](6.3.1.1(3))

$= 0.84034 \times 58.7\ cm^2 \times 355\ MPa / 1$

$= 1751.1\ kN$

Buckling resistance about y

[52] $N_{cr,z} = pi^2 * E * I_z / L_{cr,z}^2$ [1](6.3.1.2(1))

$= \pi^2 \times 210\ GPa \times 1548\ cm^4 / (4\ m)^2$

$= 2005.3\ kN$

Elastic critical force about z

[53] $\lambda_z = \sqrt{A * f_y / N_{cr,z}}$ [1](6.3.1.2(1))

$= \sqrt{58.7\ cm^2 \times 355\ MPa / 2005.3\ kN}$

$= 1.0194$

Non-dimensional slenderness about z

[54] $\Phi_z = 0.5 * (1 + \alpha_z * (\lambda_z - 0.2) + \lambda_z^2)$ [1](6.3.1.2(1))

$= 0.5 \times (1 + 0.49 \times (1.0194 - 0.2) + 1.0194^2)$

$= 1.2204$

Φ about z

[55] $\chi_z = \min(1, 1 / (\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}))$ [1](6.3.1.2(1))

$= \min(1, 1 / (1.2204 + \sqrt{1.2204^2 - 1.0194^2}))$

$= 0.52876$

Reduction factor χ about z

[56] $N_{b,z,Rd} = \chi_z * A * f_y / \gamma_{M1}$ [1](6.3.1.1(3))

$= 0.52876 \times 58.7\ cm^2 \times 355\ MPa / 1$

$= 1101.9\ kN$

Buckling resistance about z

[57] $N_{b,Rd} = \min(N_{b,y,Rd}, N_{b,z,Rd})$

$= \min(1751.1\ kN, 1101.9\ kN)$

$= 1101.9\ kN$

Governing buckling resistance

[58] $u_{buckling} = \text{abs}(N_{Ed}) / N_{b,Rd}$ [1](6.3.1.1(1))

$= \text{abs}(-900\ kN) / 1101.9\ kN$

$= 0.8168$

Flexural buckling utilisation

[59] $M_{y,b,Rd} = W_y * f_y / \gamma_{M1}$ [1](6.3.3, Eq. 6.61)

$= 497\ cm^3 \times 355\ MPa / 1$

$= 176.44\ kN \cdot m$

Characteristic bending resistance about y over $\gamma_{M1}$ (member check, no shear reduction)

[60] $M_{z,b,Rd} = W_z * f_y / \gamma_{M1}$ [1](6.3.3, Eq. 6.62)

$= 231\ cm^3 \times 355\ MPa / 1$

$= 82.005\ kN \cdot m$

Characteristic bending resistance about z over $\gamma_{M1}$ (member check, no shear reduction)

[61] $u_{member} = \text{abs}(N_{Ed}) / N_{b,Rd} + k_{yy} * M_{y,Ed} / M_{y,b,Rd} + k_{zz} * M_{z,Ed} / M_{z,b,Rd}$ [1](6.3.3(4))

$= \text{abs}(-900\ kN) / 1101.9\ kN + 1 \times 40\ kN \cdot m / 176.44\ kN \cdot m + 1 \times 0\ kN \cdot m / 82.005\ kN \cdot m$

$= 1.0435$

Simplified member interaction

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $A_v$ | 16.94 cm² | Shear area |
| $N_{pl,Rd}$ | 2083.8 kN | $N_{pl,Rd}$ |
| $V_{pl,Rd}$ | 347.28 kN | $V_{pl,Rd}$ |
| ρ | 0 | ρ |
| $M_{y,c,Rd}$ | 176.44 kN·m | $M_{y,c,Rd}$ |
| $M_{z,c,Rd}$ | 82.01 kN·m | $M_{z,c,Rd}$ |
| $M_{y,b,Rd}$ | 176.44 kN·m | $M_{y,Rk}/\gamma_{M1}$ |
| $M_{z,b,Rd}$ | 82.01 kN·m | $M_{z,Rk}/\gamma_{M1}$ |
| $N_{cr,y}$ | 5917.3 kN | $N_{cr,y}$ |
| $N_{cr,z}$ | 2005.3 kN | $N_{cr,z}$ |
| $\lambda_y$ | 0.59343 | $\bar{\lambda}_y$ |
| $\lambda_z$ | 1.0194 | $\bar{\lambda}_z$ |
| $\chi_y$ | 0.84034 | $\chi_y$ |
| $\chi_z$ | 0.52876 | $\chi_z$ |
| $N_{b,y,Rd}$ | 1751.1 kN | $N_{b,y,Rd}$ |
| $N_{b,z,Rd}$ | 1101.9 kN | $N_{b,z,Rd}$ |
| $N_{b,Rd}$ | 1101.9 kN | $N_{b,Rd}$ |
| $u_N$ | 0.43 | Axial utilisation |
| $u_{M,y}$ | 0.23 | Bending y utilisation |
| $u_{M,z}$ | 0.00 | Bending z utilisation |
| $u_V$ | 0.06 | Shear utilisation |
| $u_{section}$ | 0.66 | Cross-section interaction |
| $u_{buckling}$ | 0.82 | Buckling utilisation |
| $u_{member}$ | 1.05 | Member interaction (governing) |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | Member interaction (simplified) | $u_{member} = 1.05$ | $u_{member} \le 1$ | 1.05 | [1](6.3.3) |
| PASS | Flexural buckling $N_{Ed} \le N_{b,Rd}$ | $u_{buckling} = 0.82$ | $u_{buckling} \le 1$ | 0.82 | [1](6.3.1.1) |
| PASS | Cross-section interaction | $u_{section} = 0.66$ | $u_{section} \le 1$ | 0.66 | [1](6.2.1(7)) |
| PASS | Compression $N_{Ed} \le N_{c,Rd}$ | $u_N = 0.43$ | $u_N \le 1$ | 0.43 | [1](6.2.4) |

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Bending y $M_{y,Ed} \le M_{y,c,Rd}$ | $u_{M,y} = 0.23$ | $u_{M,y} \le 1$ | 0.23 | [1](6.2.5) |
| PASS | Shear $V_{Ed} \le V_{pl,Rd}$ | $u_{V} = 0.06$ | $u_{V} \le 1$ | 0.06 | [1](6.2.6) |
| PASS | Bending z $M_{z,Ed} \le M_{z,c,Rd}$ | $u_{M,z} = 0.00$ | $u_{M,z} \le 1$ | 0.00 | [1](6.2.5) |

## 7. Applicability

WARNING $M_{y,Ed}$: lateral-torsional buckling is not checked by this sheet; restrain the compression flange or check 6.3.2 separately. [no-ltb]
WARNING $k_{yy}$, $k_{zz}$: $k_{yy}$ is left at 1.0 although $N_{Ed}/N_{b,Rd}$ = 0.817 exceeds 0.3. Annex A or B give k above 1 here (at most $C_m(1 + 0.8n)$, which is $k_{yy}$ = 1.411 with $C_m$ = 1); with those values the member interaction would be 1.137 against 1.044 shown. Enter k from Annex A or B. [k-default]

## 8. Verification

Template struct.en1993-member@1.3.1 Content hash a2e6114eb2412c02 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
2. Gardner L, Nethercot DA. Designers' Guide to EN 1993-1-1: Eurocode 3: Design of Steel Structures, General Rules and Rules for Buildings. Thomas Telford; 2005.